

Student Management System (SMS)

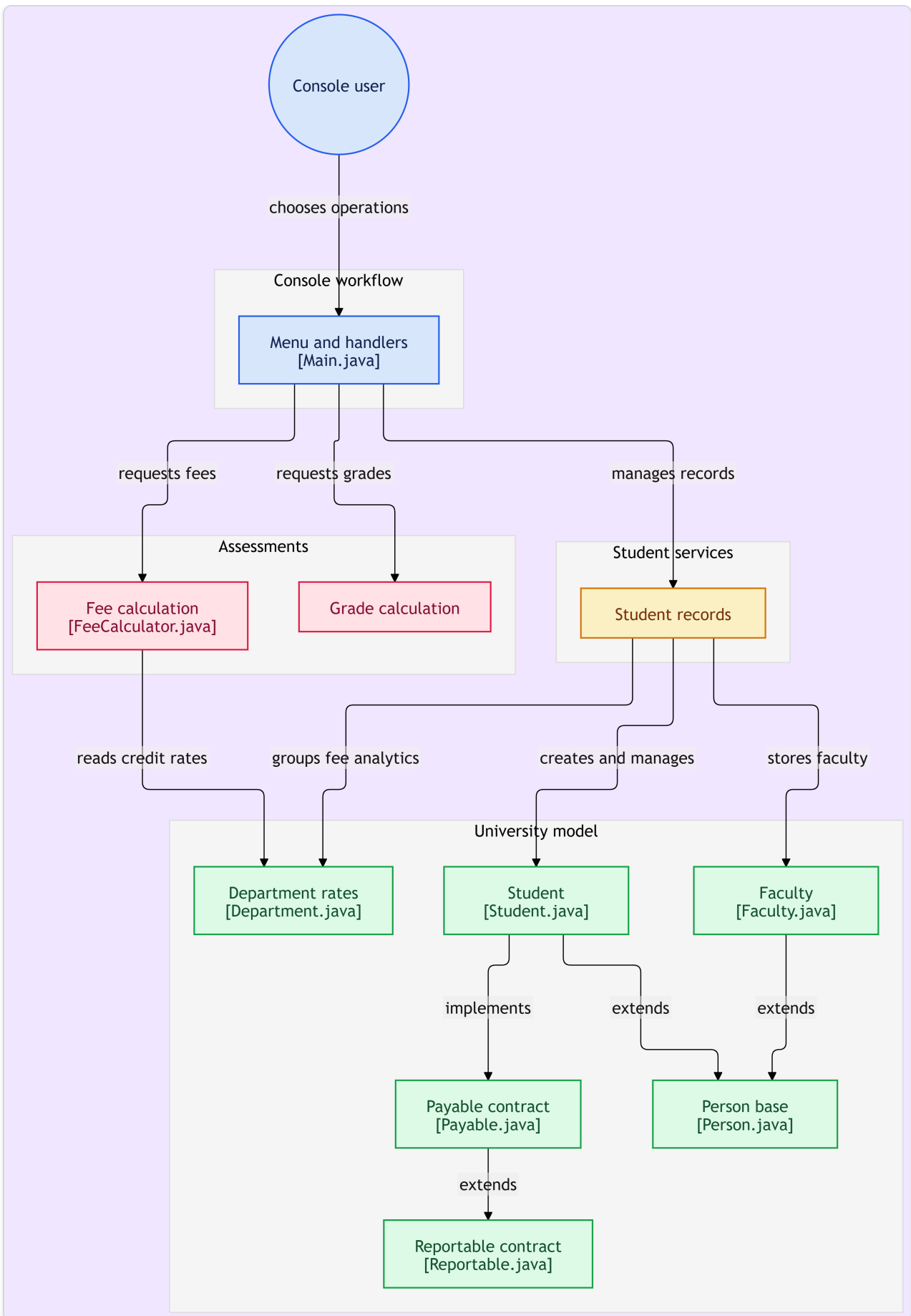
Console-based Java OOP mini project for REVA University student records, grading, fees, and syllabus demonstrations.

Field	Detail
Course	B.Sc. (BSTCs) Sem V — Java Programming (Units I & II)
Author	A Adisri Menon — SRN: R24SA001
Institution	REVA University, Bengaluru
Environment	Java SE (JDK 17+)

Features

- **Registration** — SRN, department, semester, credits, hostel, scholarship.
- **Grading** — 5-subject marks → percentage → grade (O / A / B / C / D / F).
- **Fee Engine** — tuition + hostel + amenities – scholarship discount.
- **Search & Sort** — SRN / partial-name search; bubble-sort by name, SRN, semester, fee.
- **Evaluator Demo** — 14 interactive OOP demonstrations covering all 21 rubric items.

Workflow



Output Screenshots

REVA UNIVERSITY STUDENT MANAGEMENT SYSTEM

- ```
=====
```
1. Add New Student
  2. Display All Student Records
  3. Search Student by SRN
  4. Search Student by Partial Name
  5. Calculate / Record Semester Fee
  6. Remove Student Record
  7. Sort Students
  8. View Department Fee Analytics
  9. View University Personnel Directory
  10. OOP & Java Concept Demonstrations
  0. Exit
- ```
=====
```

Enter your choice:

Main Menu

=====

Enter your choice: 1

--- [ADD NEW STUDENT] ---

Enter Student Full Name: Harry Justin

Enter Age (16 - 60): 18

Enter SRN (e.g., R24CS010): R26SA020

Select Academic Department:

1. CSE - Computer Science & Engineering
2. ECE - Electronics & Communication Engineering
3. MECH - Mechanical Engineering
4. CIVIL - Civil Engineering
5. ISE - Information Science & Engineering
6. EEE - Electrical & Electronics Engineering
7. BT - Biotechnology
8. AHS - Allied Health Sciences
9. COM - School of Commerce
10. MGT - School of Management Studies
11. PA - School of Performing Arts

Select Department (1-11): 10

Enter Current Semester (1-8): 1

Is the student residing in University Hostel? (yes/no): Y

Enter Registered Course Credits (10-30): 28

Enter Scholarship Discount Percentage (0 - 100): 80

Enter Marks for 5 Core Subjects (0 - 100 each):

Subject 1 Marks: 96

Subject 2 Marks: 98

Subject 3 Marks: 92

Subject 4 Marks: 93

Subject 5 Marks: 95

[SUCCESS] Student Record Successfully Created!

-> Assigned Student ID : REVA-STU-1034

[SUCCESS] Student Record Successfully Created!

-> Assigned Student ID : REVA-STU-1034

-> Total Marks : 474 / 500 (94.80%)

-> Computed Grade : O (Outstanding Performance (Top Tier))

-> Computed Semester Fee: INR 24140.00

----- Fee Breakdown -----

Tuition Fee : INR 67200.00 (28 credits @ 2400.00)

Campus Amenities : INR 8500.00

Hostel & Mess Charge : INR 45000.00

Gross Total : INR 120700.00

Scholarship Deduction : INR 96560.00 (80.0% discount)

Net Semester Fee : INR 24140.00

Add Student — Input & Result

Enter your choice: 2

STUDENT ID	SRN	NAME	DEPT	SEM	GRADE	FEE DUE (INR)	HOSTELLER
REVA-STU-1001	R24CS001	Arjun Nair	CSE	5	O	63975.00	NO
REVA-STU-1002	R24CS002	Priya Sharma	CSE	5	O	105315.00	YES
REVA-STU-1003	R24CS003	Rahul Menon	CSE	6	A	72500.00	NO
REVA-STU-1004	R24EC001	Sneha Iyer	ECE	5	A	107550.00	YES
REVA-STU-1005	R24EC002	Karthik Rao	ECE	6	B	68500.00	NO
REVA-STU-1006	R24EC003	Ananya Krishnan	ECE	3	O	100400.00	YES
REVA-STU-1007	R24ME001	Rohan Mehta	MECH	7	B	98325.00	YES
REVA-STU-1008	R24ME002	Meera Nair	MECH	4	A	57150.00	NO
REVA-STU-1009	R24ME003	Vivek Kumar	MECH	5	C	98500.00	YES
REVA-STU-1010	R24CV001	Aditi Shah	CIVIL	4	A	56500.00	NO
REVA-STU-1011	R24CV002	Nikhil Varma	CIVIL	6	B	95670.00	YES
REVA-STU-1012	R24CV003	Diya Reddy	CIVIL	3	A	48025.00	NO
REVA-STU-1013	R24IS001	Harish Bhat	ISE	5	O	61360.00	NO
REVA-STU-1014	R24IS002	Kavya Rao	ISE	5	A	103950.00	YES
REVA-STU-1015	R24IS003	Akash Jain	ISE	4	D	70500.00	NO
REVA-STU-1016	R24EE001	Neha Pillai	EEE	6	A	97835.00	YES
REVA-STU-1017	R24EE002	Rohit Das	EEE	4	B	64500.00	NO
REVA-STU-1018	R24EE003	Ishita Menon	EEE	3	O	90525.00	YES
REVA-STU-1019	R24BT001	Varun Shetty	BT	4	A	61110.00	NO
REVA-STU-1020	R24BT002	Pooja Nair	BT	5	O	94640.00	YES
REVA-STU-1021	R24BT003	Manoj Kulkarni	BT	4	C	57100.00	NO
REVA-STU-1022	R24AH001	Shreya Joshi	AHS	4	A	99630.00	YES
REVA-STU-1023	R24AH002	Deepa Hegde	AHS	5	A	51425.00	NO
REVA-STU-1024	R24AH003	Tanvi Deshmukh	AHS	2	B	100300.00	YES
REVA-STU-1025	R24CM001	Aditya Verma	COM	4	A	52500.00	NO
REVA-STU-1026	R24CM002	Bhavana Murthy	COM	6	O	81520.00	YES
REVA-STU-1027	R24CM003	Chetan Gowda	COM	4	C	48100.00	NO
REVA-STU-1028	R24MG001	Divya Swaminathan	MGT	5	O	94435.00	YES
REVA-STU-1029	R24MG002	Eashan Prabhu	MGT	4	B	56500.00	NO
REVA-STU-1030	R24MG003	Fatima Zohra	MGT	6	A	95670.00	YES
REVA-STU-1031	R24PA001	Gayatri Nambiar	PA	4	O	39375.00	NO
REVA-STU-1032	R24PA002	Hemant Shastri	PA	5	B	84150.00	YES
REVA-STU-1033	R24PA003	Indu Ramaswamy	PA	3	A	44500.00	NO
REVA-STU-1034	R26SA020	Harry Justin	MGT	1	O	24140.00	YES

Display All Students

```
=====
REVA UNIVERSITY STUDENT MANAGEMENT SYSTEM
=====
```

1. Add New Student
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0. Exit

```
=====
Enter your choice: 3
```

```
--- [SEARCH BY SRN] ---
```

```
Enter SRN to search (e.g. R24CS001): R24AH003
```

```
[MATCH FOUND]:
```

```
-----
[STUDENT RECORD] Tanvi Deshmukh (SRN: R24AH003)
-----
```

Student ID	: REVA-STU-1024
Full Name	: Tanvi Deshmukh
Age	: 19
SRN	: R24AH003
Department	: Allied Health Sciences
Semester	: 2
Admission Number	: 2026000024
Hosteller	: YES
Scholarship	: 0.0%
Marks (5 Subjects)	: [72, 70, 75, 68, 71]
Calculated Grade	: B
Outstanding Fee	: INR 100300.00

Search by SRN

REVA UNIVERSITY STUDENT MANAGEMENT SYSTEM

- ```
=====
```
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  0. Exit

```
=====
```

Enter your choice: 4

--- [SEARCH BY PARTIAL NAME] ---

Enter name or partial letters: Shet

[FOUND 1 MATCHING RECORD(S)]:

```

```

[STUDENT RECORD] Varun Shetty (SRN: R24BT001)

```

```

|                    |                        |
|--------------------|------------------------|
| Student ID         | : REVA-STU-1019        |
| Full Name          | : Varun Shetty         |
| Age                | : 20                   |
| SRN                | : R24BT001             |
| Department         | : Biotechnology        |
| Semester           | : 4                    |
| Admission Number   | : 2026000019           |
| Hosteller          | : NO                   |
| Scholarship        | : 10.0%                |
| Marks (5 Subjects) | : [88, 85, 90, 86, 87] |
| Calculated Grade   | : A                    |
| Outstanding Fee    | : INR 61110.00         |

```

```

=====

Enter your choice: 5

-----

CALCULATE / RECORD SEMESTER FEE

-----

Enter Student SRN: R24BT001

----- Student Fee Assessment -----

Student : Varun Shetty  
SRN : R24BT001  
Department : Biotechnology  
Semester : 4  
Credits : 22 (Base Rate: INR 2700.00/credit)  
Hosteller : NO  
Scholarship : 10.0%  
Semester Fee : INR 61110.00  
Current Due : INR 61110.00

-----

Enter Payment Amount (INR): 60000

[SUCCESS] Payment of INR 60000.00 recorded for Varun Shetty (R24BT001). New Balance: INR 1110.00

-----

FEE PAYMENT RECEIPT

-----

Student : Varun Shetty  
SRN : R24BT001  
Total Fee : INR 61110.00  
Previous Due : INR 61110.00  
Payment : INR 60000.00  
Remaining Due : INR 1110.00

-----

Fee Payment

```
=====
REVA UNIVERSITY STUDENT MANAGEMENT SYSTEM
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```
=====
Enter your choice: 6
```

```
--- [REMOVE STUDENT RECORD] ---
```

```
Enter SRN of student to remove: R26SA020
```

```
Confirm removal of 'Harry Justin' (SRN: R26SA020)?
```

```
Type 'yes' to confirm or 'no' to cancel: Y
```

```
[SUCCESS] Student record 'R26SA020' removed from database.
```

Remove Student

Enter your choice: 7

--- [SORT STUDENT RECORDS] ---

Select Sorting Criteria:

1. Sort by Student Name (Alphabetical)
2. Sort by Student SRN
3. Sort by Semester
4. Sort by Fee Due (Ascending)
0. Return to Main Menu

Enter sorting choice (0-4): 1

[SUCCESS] Records sorted alphabetically by Student Name.

| STUDENT ID    | SRN      | NAME              | DEPT  | SEM | GRADE | FEE DUE (INR) | HOSTELLER |
|---------------|----------|-------------------|-------|-----|-------|---------------|-----------|
| REVA-STU-1010 | R24CV001 | Aditi Shah        | CIVIL | 4   | A     | 56500.00      | NO        |
| REVA-STU-1025 | R24CM001 | Aditya Verma      | COM   | 4   | A     | 52500.00      | NO        |
| REVA-STU-1015 | R24IS003 | Akash Jain        | ISE   | 4   | D     | 70500.00      | NO        |
| REVA-STU-1006 | R24EC003 | Ananya Krishnan   | ECE   | 3   | O     | 100400.00     | YES       |
| REVA-STU-1001 | R24CS001 | Arjun Nair        | CSE   | 5   | O     | 63975.00      | NO        |
| REVA-STU-1026 | R24CM002 | Bhavana Murthy    | COM   | 6   | O     | 81520.00      | YES       |
| REVA-STU-1027 | R24CM003 | Chetan Gowda      | COM   | 4   | C     | 48100.00      | NO        |
| REVA-STU-1023 | R24AH002 | Deepa Hegde       | AHS   | 5   | A     | 51425.00      | NO        |
| REVA-STU-1028 | R24MG001 | Divya Swaminathan | MGT   | 5   | O     | 94435.00      | YES       |
| REVA-STU-1012 | R24CV003 | Diya Reddy        | CIVIL | 3   | A     | 48025.00      | NO        |
| REVA-STU-1029 | R24MG002 | Eashan Prabhu     | MGT   | 4   | B     | 56500.00      | NO        |
| REVA-STU-1030 | R24MG003 | Fatima Zohra      | MGT   | 6   | A     | 95670.00      | YES       |
| REVA-STU-1031 | R24PA001 | Gayatri Nambiar   | PA    | 4   | O     | 39375.00      | NO        |
| REVA-STU-1013 | R24IS001 | Harish Bhat       | ISE   | 5   | O     | 61360.00      | NO        |
| REVA-STU-1032 | R24PA002 | Hemant Shastri    | PA    | 5   | B     | 84150.00      | YES       |
| REVA-STU-1033 | R24PA003 | Indu Ramaswamy    | PA    | 3   | A     | 44500.00      | NO        |
| REVA-STU-1018 | R24EE003 | Ishita Menon      | EEE   | 3   | O     | 90525.00      | YES       |
| REVA-STU-1005 | R24EC002 | Karthik Rao       | ECE   | 6   | B     | 68500.00      | NO        |
| REVA-STU-1014 | R24IS002 | Kavya Rao         | ISE   | 5   | A     | 103950.00     | YES       |
| REVA-STU-1021 | R24BT003 | Manoj Kulkarni    | BT    | 4   | C     | 57100.00      | NO        |
| REVA-STU-1008 | R24ME002 | Meera Nair        | MECH  | 4   | A     | 57150.00      | NO        |
| REVA-STU-1016 | R24EE001 | Neha Pillai       | EEE   | 6   | A     | 97835.00      | YES       |
| REVA-STU-1011 | R24CV002 | Nikhil Varma      | CIVIL | 6   | B     | 95670.00      | YES       |

|               |          |                |       |   |   |           |     |
|---------------|----------|----------------|-------|---|---|-----------|-----|
| REVA-STU-1033 | R24PA003 | Indu Ramaswamy | PA    | 3 | A | 44500.00  | NO  |
| REVA-STU-1018 | R24EE003 | Ishita Menon   | EEE   | 3 | O | 90525.00  | YES |
| REVA-STU-1005 | R24EC002 | Karthik Rao    | ECE   | 6 | B | 68500.00  | NO  |
| REVA-STU-1014 | R24IS002 | Kavya Rao      | ISE   | 5 | A | 103950.00 | YES |
| REVA-STU-1021 | R24BT003 | Manoj Kulkarni | BT    | 4 | C | 57100.00  | NO  |
| REVA-STU-1008 | R24ME002 | Meera Nair     | MECH  | 4 | A | 57150.00  | NO  |
| REVA-STU-1016 | R24EE001 | Neha Pillai    | EEE   | 6 | A | 97835.00  | YES |
| REVA-STU-1011 | R24CV002 | Nikhil Varma   | CIVIL | 6 | B | 95670.00  | YES |
| REVA-STU-1020 | R24BT002 | Pooja Nair     | BT    | 5 | O | 94640.00  | YES |
| REVA-STU-1002 | R24CS002 | Priya Sharma   | CSE   | 5 | O | 105315.00 | YES |
| REVA-STU-1003 | R24CS003 | Rahul Menon    | CSE   | 6 | A | 72500.00  | NO  |
| REVA-STU-1007 | R24ME001 | Rohan Mehta    | MECH  | 7 | B | 98325.00  | YES |
| REVA-STU-1017 | R24EE002 | Rohit Das      | EEE   | 4 | B | 64500.00  | NO  |
| REVA-STU-1022 | R24AH001 | Shreya Joshi   | AHS   | 4 | A | 99630.00  | YES |
| REVA-STU-1004 | R24EC001 | Sneha Iyer     | ECE   | 5 | A | 107550.00 | YES |
| REVA-STU-1024 | R24AH003 | Tanvi Deshmukh | AHS   | 2 | B | 100300.00 | YES |
| REVA-STU-1019 | R24BT001 | Varun Shetty   | BT    | 4 | A | 1110.00   | NO  |
| REVA-STU-1009 | R24ME003 | Vivek Kumar    | MECH  | 5 | C | 98500.00  | YES |

=====

Total Enrolled Students: 33 | Total Database Capacity: 100

=====

REVA UNIVERSITY STUDENT MANAGEMENT SYSTEM

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0. Exit

=====

Enter your choice:

Sort by Name

Enter your choice: 7

--- [SORT STUDENT RECORDS] ---

Select Sorting Criteria:

1. Sort by Student Name (Alphabetical)
2. Sort by Student SRN
3. Sort by Semester
4. Sort by Fee Due (Ascending)
0. Return to Main Menu

Enter sorting choice (0-4): 2

[SUCCESS] Records sorted by Student SRN.

| STUDENT ID    | SRN      | NAME            | DEPT  | SEM | GRADE | FEE DUE (INR) | HOSTELLER |
|---------------|----------|-----------------|-------|-----|-------|---------------|-----------|
| REVA-STU-1022 | R24AH001 | Shreya Joshi    | AHS   | 4   | A     | 99630.00      | YES       |
| REVA-STU-1023 | R24AH002 | Deepa Hegde     | AHS   | 5   | A     | 51425.00      | NO        |
| REVA-STU-1024 | R24AH003 | Tanvi Deshmukh  | AHS   | 2   | B     | 100300.00     | YES       |
| REVA-STU-1019 | R24BT001 | Varun Shetty    | BT    | 4   | A     | 1110.00       | NO        |
| REVA-STU-1020 | R24BT002 | Pooja Nair      | BT    | 5   | O     | 94640.00      | YES       |
| REVA-STU-1021 | R24BT003 | Manoj Kulkarni  | BT    | 4   | C     | 57100.00      | NO        |
| REVA-STU-1025 | R24CM001 | Aditya Verma    | COM   | 4   | A     | 52500.00      | NO        |
| REVA-STU-1026 | R24CM002 | Bhavana Murthy  | COM   | 6   | O     | 81520.00      | YES       |
| REVA-STU-1027 | R24CM003 | Chetan Gowda    | COM   | 4   | C     | 48100.00      | NO        |
| REVA-STU-1001 | R24CS001 | Arjun Nair      | CSE   | 5   | O     | 63975.00      | NO        |
| REVA-STU-1002 | R24CS002 | Priya Sharma    | CSE   | 5   | O     | 105315.00     | YES       |
| REVA-STU-1003 | R24CS003 | Rahul Menon     | CSE   | 6   | A     | 72500.00      | NO        |
| REVA-STU-1010 | R24CV001 | Aditi Shah      | CIVIL | 4   | A     | 56500.00      | NO        |
| REVA-STU-1011 | R24CV002 | Nikhil Varma    | CIVIL | 6   | B     | 95670.00      | YES       |
| REVA-STU-1012 | R24CV003 | Diya Reddy      | CIVIL | 3   | A     | 48025.00      | NO        |
| REVA-STU-1004 | R24EC001 | Sneha Iyer      | ECE   | 5   | A     | 107550.00     | YES       |
| REVA-STU-1005 | R24EC002 | Karthik Rao     | ECE   | 6   | B     | 68500.00      | NO        |
| REVA-STU-1006 | R24EC003 | Ananya Krishnan | ECE   | 3   | O     | 100400.00     | YES       |
| REVA-STU-1016 | R24EE001 | Neha Pillai     | EEE   | 6   | A     | 97835.00      | YES       |
| REVA-STU-1017 | R24EE002 | Rohit Das       | EEE   | 4   | B     | 64500.00      | NO        |
| REVA-STU-1018 | R24EE003 | Ishita Menon    | EEE   | 3   | O     | 90525.00      | YES       |
| REVA-STU-1013 | R24IS001 | Harish Bhat     | ISE   | 5   | O     | 61360.00      | NO        |
| REVA-STU-1014 | R24IS002 | Kavya Rao       | ISE   | 5   | A     | 103950.00     | YES       |

|               |          |                   |      |   |   |           |     |
|---------------|----------|-------------------|------|---|---|-----------|-----|
| REVA-STU-1004 | R24EC001 | Sneha Iyer        | ECE  | 5 | A | 107550.00 | YES |
| REVA-STU-1005 | R24EC002 | Karthik Rao       | ECE  | 6 | B | 68500.00  | NO  |
| REVA-STU-1006 | R24EC003 | Ananya Krishnan   | ECE  | 3 | O | 100400.00 | YES |
| REVA-STU-1016 | R24EE001 | Neha Pillai       | EEE  | 6 | A | 97835.00  | YES |
| REVA-STU-1017 | R24EE002 | Rohit Das         | EEE  | 4 | B | 64500.00  | NO  |
| REVA-STU-1018 | R24EE003 | Ishita Menon      | EEE  | 3 | O | 90525.00  | YES |
| REVA-STU-1013 | R24IS001 | Harish Bhat       | ISE  | 5 | O | 61360.00  | NO  |
| REVA-STU-1014 | R24IS002 | Kavya Rao         | ISE  | 5 | A | 103950.00 | YES |
| REVA-STU-1015 | R24IS003 | Akash Jain        | ISE  | 4 | D | 70500.00  | NO  |
| REVA-STU-1007 | R24ME001 | Rohan Mehta       | MECH | 7 | B | 98325.00  | YES |
| REVA-STU-1008 | R24ME002 | Meera Nair        | MECH | 4 | A | 57150.00  | NO  |
| REVA-STU-1009 | R24ME003 | Vivek Kumar       | MECH | 5 | C | 98500.00  | YES |
| REVA-STU-1028 | R24MG001 | Divya Swaminathan | MGT  | 5 | O | 94435.00  | YES |
| REVA-STU-1029 | R24MG002 | Eashan Prabhu     | MGT  | 4 | B | 56500.00  | NO  |
| REVA-STU-1030 | R24MG003 | Fatima Zohra      | MGT  | 6 | A | 95670.00  | YES |
| REVA-STU-1031 | R24PA001 | Gayatri Nambiar   | PA   | 4 | O | 39375.00  | NO  |
| REVA-STU-1032 | R24PA002 | Hemant Shastri    | PA   | 5 | B | 84150.00  | YES |
| REVA-STU-1033 | R24PA003 | Indu Ramaswamy    | PA   | 3 | A | 44500.00  | NO  |

=====  
Total Enrolled Students: 33 | Total Database Capacity: 100

=====

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=====  
Enter your choice: █

Sort by SRN

Enter your choice: 7

--- [SORT STUDENT RECORDS] ---

Select Sorting Criteria:

1. Sort by Student Name (Alphabetical)
2. Sort by Student SRN
3. Sort by Semester
4. Sort by Fee Due (Ascending)
0. Return to Main Menu

Enter sorting choice (0-4): 3

[SUCCESS] Records sorted by Semester.

| STUDENT ID    | SRN      | NAME            | DEPT  | SEM | GRADE | FEE DUE (INR) | HOSTELLER |
|---------------|----------|-----------------|-------|-----|-------|---------------|-----------|
| REVA-STU-1024 | R24AH003 | Tanvi Deshmukh  | AHS   | 2   | B     | 100300.00     | YES       |
| REVA-STU-1012 | R24CV003 | Diya Reddy      | CIVIL | 3   | A     | 48025.00      | NO        |
| REVA-STU-1006 | R24EC003 | Ananya Krishnan | ECE   | 3   | O     | 100400.00     | YES       |
| REVA-STU-1018 | R24EE003 | Ishita Menon    | EEE   | 3   | O     | 90525.00      | YES       |
| REVA-STU-1033 | R24PA003 | Indu Ramaswamy  | PA    | 3   | A     | 44500.00      | NO        |
| REVA-STU-1022 | R24AH001 | Shreya Joshi    | AHS   | 4   | A     | 99630.00      | YES       |
| REVA-STU-1019 | R24BT001 | Varun Shetty    | BT    | 4   | A     | 1110.00       | NO        |
| REVA-STU-1021 | R24BT003 | Manoj Kulkarni  | BT    | 4   | C     | 57100.00      | NO        |
| REVA-STU-1025 | R24CM001 | Aditya Verma    | COM   | 4   | A     | 52500.00      | NO        |
| REVA-STU-1027 | R24CM003 | Chetan Gowda    | COM   | 4   | C     | 48100.00      | NO        |
| REVA-STU-1010 | R24CV001 | Aditi Shah      | CIVIL | 4   | A     | 56500.00      | NO        |
| REVA-STU-1017 | R24EE002 | Rohit Das       | EEE   | 4   | B     | 64500.00      | NO        |
| REVA-STU-1015 | R24IS003 | Akash Jain      | ISE   | 4   | D     | 70500.00      | NO        |
| REVA-STU-1008 | R24ME002 | Meera Nair      | MECH  | 4   | A     | 57150.00      | NO        |
| REVA-STU-1029 | R24MG002 | Eashan Prabhu   | MGT   | 4   | B     | 56500.00      | NO        |
| REVA-STU-1031 | R24PA001 | Gayatri Nambiar | PA    | 4   | O     | 39375.00      | NO        |
| REVA-STU-1023 | R24AH002 | Deepa Hegde     | AHS   | 5   | A     | 51425.00      | NO        |
| REVA-STU-1020 | R24BT002 | Pooja Nair      | BT    | 5   | O     | 94640.00      | YES       |
| REVA-STU-1001 | R24CS001 | Arjun Nair      | CSE   | 5   | O     | 63975.00      | NO        |
| REVA-STU-1002 | R24CS002 | Priya Sharma    | CSE   | 5   | O     | 105315.00     | YES       |
| REVA-STU-1004 | R24EC001 | Sneha Iyer      | ECE   | 5   | A     | 107550.00     | YES       |
| REVA-STU-1013 | R24IS001 | Harish Bhat     | ISE   | 5   | O     | 61360.00      | NO        |
| REVA-STU-1014 | R24IS002 | Kavya Rao       | ISE   | 5   | A     | 103950.00     | YES       |

|               |          |                   |       |   |   |           |     |
|---------------|----------|-------------------|-------|---|---|-----------|-----|
| REVA-STU-1031 | R24PA001 | Gayatri Nambiar   | PA    | 4 | O | 39375.00  | NO  |
| REVA-STU-1023 | R24AH002 | Deepa Hegde       | AHS   | 5 | A | 51425.00  | NO  |
| REVA-STU-1020 | R24BT002 | Pooja Nair        | BT    | 5 | O | 94640.00  | YES |
| REVA-STU-1001 | R24CS001 | Arjun Nair        | CSE   | 5 | O | 63975.00  | NO  |
| REVA-STU-1002 | R24CS002 | Priya Sharma      | CSE   | 5 | O | 105315.00 | YES |
| REVA-STU-1004 | R24EC001 | Sneha Iyer        | ECE   | 5 | A | 107550.00 | YES |
| REVA-STU-1013 | R24IS001 | Harish Bhat       | ISE   | 5 | O | 61360.00  | NO  |
| REVA-STU-1014 | R24IS002 | Kavya Rao         | ISE   | 5 | A | 103950.00 | YES |
| REVA-STU-1009 | R24ME003 | Vivek Kumar       | MECH  | 5 | C | 98500.00  | YES |
| REVA-STU-1028 | R24MG001 | Divya Swaminathan | MGT   | 5 | O | 94435.00  | YES |
| REVA-STU-1032 | R24PA002 | Hemant Shastri    | PA    | 5 | B | 84150.00  | YES |
| REVA-STU-1026 | R24CM002 | Bhavana Murthy    | COM   | 6 | O | 81520.00  | YES |
| REVA-STU-1003 | R24CS003 | Rahul Menon       | CSE   | 6 | A | 72500.00  | NO  |
| REVA-STU-1011 | R24CV002 | Nikhil Varma      | CIVIL | 6 | B | 95670.00  | YES |
| REVA-STU-1005 | R24EC002 | Karthik Rao       | ECE   | 6 | B | 68500.00  | NO  |
| REVA-STU-1016 | R24EE001 | Neha Pillai       | EEE   | 6 | A | 97835.00  | YES |
| REVA-STU-1030 | R24MG003 | Fatima Zohra      | MGT   | 6 | A | 95670.00  | YES |
| REVA-STU-1007 | R24ME001 | Rohan Mehta       | MECH  | 7 | B | 98325.00  | YES |

Total Enrolled Students: 33 | Total Database Capacity: 100

Sort by Semester

Enter your choice: 7

--- [SORT STUDENT RECORDS] ---

Select Sorting Criteria:

1. Sort by Student Name (Alphabetical)
2. Sort by Student SRN
3. Sort by Semester
4. Sort by Fee Due (Ascending)
0. Return to Main Menu

Enter sorting choice (0-4): 4

[SUCCESS] Records sorted by Fee Due (Ascending).

| STUDENT ID    | SRN      | NAME              | DEPT  | SEM | GRADE | FEE DUE (INR) | HOSTELLER |
|---------------|----------|-------------------|-------|-----|-------|---------------|-----------|
| REVA-STU-1019 | R24BT001 | Varun Shetty      | BT    | 4   | A     | 1110.00       | NO        |
| REVA-STU-1031 | R24PA001 | Gayatri Nambiar   | PA    | 4   | O     | 39375.00      | NO        |
| REVA-STU-1033 | R24PA003 | Indu Ramaswamy    | PA    | 3   | A     | 44500.00      | NO        |
| REVA-STU-1012 | R24CV003 | Diya Reddy        | CIVIL | 3   | A     | 48025.00      | NO        |
| REVA-STU-1027 | R24CM003 | Chetan Gowda      | COM   | 4   | C     | 48100.00      | NO        |
| REVA-STU-1023 | R24AH002 | Deepa Hegde       | AHS   | 5   | A     | 51425.00      | NO        |
| REVA-STU-1025 | R24CM001 | Aditya Verma      | COM   | 4   | A     | 52500.00      | NO        |
| REVA-STU-1010 | R24CV001 | Aditi Shah        | CIVIL | 4   | A     | 56500.00      | NO        |
| REVA-STU-1029 | R24MG002 | Eashan Prabhu     | MGT   | 4   | B     | 56500.00      | NO        |
| REVA-STU-1021 | R24BT003 | Manoj Kulkarni    | BT    | 4   | C     | 57100.00      | NO        |
| REVA-STU-1008 | R24ME002 | Meera Nair        | MECH  | 4   | A     | 57150.00      | NO        |
| REVA-STU-1013 | R24IS001 | Harish Bhat       | ISE   | 5   | O     | 61360.00      | NO        |
| REVA-STU-1001 | R24CS001 | Arjun Nair        | CSE   | 5   | O     | 63975.00      | NO        |
| REVA-STU-1017 | R24EE002 | Rohit Das         | EEE   | 4   | B     | 64500.00      | NO        |
| REVA-STU-1005 | R24EC002 | Karthik Rao       | ECE   | 6   | B     | 68500.00      | NO        |
| REVA-STU-1015 | R24IS003 | Akash Jain        | ISE   | 4   | D     | 70500.00      | NO        |
| REVA-STU-1003 | R24CS003 | Rahul Menon       | CSE   | 6   | A     | 72500.00      | NO        |
| REVA-STU-1026 | R24CM002 | Bhavana Murthy    | COM   | 6   | O     | 81520.00      | YES       |
| REVA-STU-1032 | R24PA002 | Hemant Shastri    | PA    | 5   | B     | 84150.00      | YES       |
| REVA-STU-1018 | R24EE003 | Ishita Menon      | EEE   | 3   | O     | 90525.00      | YES       |
| REVA-STU-1028 | R24MG001 | Divya Swaminathan | MGT   | 5   | O     | 94435.00      | YES       |
| REVA-STU-1020 | R24BT002 | Pooja Nair        | BT    | 5   | O     | 94640.00      | YES       |
| REVA-STU-1011 | R24CV002 | Nikhil Varma      | CIVIL | 6   | B     | 95670.00      | YES       |

|               |          |                   |       |   |   |           |     |
|---------------|----------|-------------------|-------|---|---|-----------|-----|
| REVA-STU-1015 | R24IS003 | Akash Jain        | ISE   | 4 | D | 70500.00  | NO  |
| REVA-STU-1003 | R24CS003 | Rahul Menon       | CSE   | 6 | A | 72500.00  | NO  |
| REVA-STU-1026 | R24CM002 | Bhavana Murthy    | COM   | 6 | O | 81520.00  | YES |
| REVA-STU-1032 | R24PA002 | Hemant Shastri    | PA    | 5 | B | 84150.00  | YES |
| REVA-STU-1018 | R24EE003 | Ishita Menon      | EEE   | 3 | O | 90525.00  | YES |
| REVA-STU-1028 | R24MG001 | Divya Swaminathan | MGT   | 5 | O | 94435.00  | YES |
| REVA-STU-1020 | R24BT002 | Pooja Nair        | BT    | 5 | O | 94640.00  | YES |
| REVA-STU-1011 | R24CV002 | Nikhil Varma      | CIVIL | 6 | B | 95670.00  | YES |
| REVA-STU-1030 | R24MG003 | Fatima Zohra      | MGT   | 6 | A | 95670.00  | YES |
| REVA-STU-1016 | R24EE001 | Neha Pillai       | EEE   | 6 | A | 97835.00  | YES |
| REVA-STU-1007 | R24ME001 | Rohan Mehta       | MECH  | 7 | B | 98325.00  | YES |
| REVA-STU-1009 | R24ME003 | Vivek Kumar       | MECH  | 5 | C | 98500.00  | YES |
| REVA-STU-1022 | R24AH001 | Shreya Joshi      | AHS   | 4 | A | 99630.00  | YES |
| REVA-STU-1024 | R24AH003 | Tanvi Deshmukh    | AHS   | 2 | B | 100300.00 | YES |
| REVA-STU-1006 | R24EC003 | Ananya Krishnan   | ECE   | 3 | O | 100400.00 | YES |
| REVA-STU-1014 | R24IS002 | Kavya Rao         | ISE   | 5 | A | 103950.00 | YES |
| REVA-STU-1002 | R24CS002 | Priya Sharma      | CSE   | 5 | O | 105315.00 | YES |
| REVA-STU-1004 | R24EC001 | Sneha Iyer        | ECE   | 5 | A | 107550.00 | YES |

Total Enrolled Students: 33 | Total Database Capacity: 100

Sort by Fee Due

Enter your choice: 8

DEPARTMENT-WISE FEE ANALYTICS & REVENUE SUMMARY

| CODE  | DEPARTMENT                              | STUDENTS | TOTAL DUE (INR) | AVERAGE DUE (INR) |
|-------|-----------------------------------------|----------|-----------------|-------------------|
| CSE   | Computer Science & Engineering          | 3        | 241790.00       | 80596.67          |
| ECE   | Electronics & Communication Engineering | 3        | 276450.00       | 92150.00          |
| MECH  | Mechanical Engineering                  | 3        | 253975.00       | 84658.33          |
| CIVIL | Civil Engineering                       | 3        | 200195.00       | 66731.67          |
| ISE   | Information Science & Engineering       | 3        | 235810.00       | 78603.33          |
| EEE   | Electrical & Electronics Engineering    | 3        | 252860.00       | 84286.67          |
| BT    | Biotechnology                           | 3        | 152850.00       | 50950.00          |
| AHS   | Allied Health Sciences                  | 3        | 251355.00       | 83785.00          |
| COM   | School of Commerce                      | 3        | 182120.00       | 60706.67          |
| MGT   | School of Management Studies            | 3        | 246605.00       | 82201.67          |
| PA    | School of Performing Arts               | 3        | 168025.00       | 56008.33          |

Total Enrolled Students Across University : 33  
Students with Cleared Dues : 0  
Students with Outstanding Dues : 33  
Gross Revenue Outstanding Across Depts : INR 2462035.00  
University Average Fee Due Per Student : INR 74607.12

Fee Analytics

=====

Enter your choice: 9

=====

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UNIVERSITY PERSONNEL DIRECTORY

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STUDENTS (33 Registered)

-----

|         |                   |               |        |          |
|---------|-------------------|---------------|--------|----------|
| [BT ]   | Varun Shetty      | SRN: R24BT001 | Sem: 4 | Grade: A |
| [PA ]   | Gayatri Nambiar   | SRN: R24PA001 | Sem: 4 | Grade: O |
| [PA ]   | Indu Ramaswamy    | SRN: R24PA003 | Sem: 3 | Grade: A |
| [CIVIL] | Diya Reddy        | SRN: R24CV003 | Sem: 3 | Grade: A |
| [COM ]  | Chetan Gowda      | SRN: R24CM003 | Sem: 4 | Grade: C |
| [AHS ]  | Deepa Hegde       | SRN: R24AH002 | Sem: 5 | Grade: A |
| [COM ]  | Aditya Verma      | SRN: R24CM001 | Sem: 4 | Grade: A |
| [CIVIL] | Aditi Shah        | SRN: R24CV001 | Sem: 4 | Grade: A |
| [MGT ]  | Eashan Prabhu     | SRN: R24MG002 | Sem: 4 | Grade: B |
| [BT ]   | Manoj Kulkarni    | SRN: R24BT003 | Sem: 4 | Grade: C |
| [MECH]  | Meera Nair        | SRN: R24ME002 | Sem: 4 | Grade: A |
| [ISE ]  | Harish Bhat       | SRN: R24IS001 | Sem: 5 | Grade: O |
| [CSE ]  | Arjun Nair        | SRN: R24CS001 | Sem: 5 | Grade: O |
| [EEE ]  | Rohit Das         | SRN: R24EE002 | Sem: 4 | Grade: B |
| [ECE ]  | Karthik Rao       | SRN: R24EC002 | Sem: 6 | Grade: B |
| [ISE ]  | Akash Jain        | SRN: R24IS003 | Sem: 4 | Grade: D |
| [CSE ]  | Rahul Menon       | SRN: R24CS003 | Sem: 6 | Grade: A |
| [COM ]  | Bhavana Murthy    | SRN: R24CM002 | Sem: 6 | Grade: O |
| [PA ]   | Hemant Shastri    | SRN: R24PA002 | Sem: 5 | Grade: B |
| [EEE ]  | Ishita Menon      | SRN: R24EE003 | Sem: 3 | Grade: O |
| [MGT ]  | Divya Swaminathan | SRN: R24MG001 | Sem: 5 | Grade: O |
| [BT ]   | Pooja Nair        | SRN: R24BT002 | Sem: 5 | Grade: O |
| [CIVIL] | Nikhil Varma      | SRN: R24CV002 | Sem: 6 | Grade: B |
| [MGT ]  | Fatima Zohra      | SRN: R24MG003 | Sem: 6 | Grade: A |
| [EEE ]  | Neha Pillai       | SRN: R24EE001 | Sem: 6 | Grade: A |
| [MECH]  | Rohan Mehta       | SRN: R24ME001 | Sem: 7 | Grade: B |
| [MECH]  | Vivek Kumar       | SRN: R24ME003 | Sem: 5 | Grade: C |
| [AHS ]  | Shreya Joshi      | SRN: R24AH001 | Sem: 4 | Grade: A |
| [AHS ]  | Tanvi Deshmukh    | SRN: R24AH003 | Sem: 2 | Grade: B |
| [ECE ]  | Ananya Krishnan   | SRN: R24EC003 | Sem: 3 | Grade: O |

|                        |                     |                |                     |                                                    |
|------------------------|---------------------|----------------|---------------------|----------------------------------------------------|
| [CSE ]                 | Arjun Nair          | SRN: R24CS001  | Sem: 5              | Grade: O                                           |
| [EEE ]                 | Rohit Das           | SRN: R24EE002  | Sem: 4              | Grade: B                                           |
| [ECE ]                 | Karthik Rao         | SRN: R24EC002  | Sem: 6              | Grade: B                                           |
| [ISE ]                 | Akash Jain          | SRN: R24IS003  | Sem: 4              | Grade: D                                           |
| [CSE ]                 | Rahul Menon         | SRN: R24CS003  | Sem: 6              | Grade: A                                           |
| [COM ]                 | Bhavana Murthy      | SRN: R24CM002  | Sem: 6              | Grade: O                                           |
| [PA ]                  | Hemant Shastri      | SRN: R24PA002  | Sem: 5              | Grade: B                                           |
| [EEE ]                 | Ishita Menon        | SRN: R24EE003  | Sem: 3              | Grade: O                                           |
| [MGT ]                 | Divya Swaminathan   | SRN: R24MG001  | Sem: 5              | Grade: O                                           |
| [BT ]                  | Pooja Nair          | SRN: R24BT002  | Sem: 5              | Grade: O                                           |
| [CIVIL]                | Nikhil Varma        | SRN: R24CV002  | Sem: 6              | Grade: B                                           |
| [MGT ]                 | Fatima Zohra        | SRN: R24MG003  | Sem: 6              | Grade: A                                           |
| [EEE ]                 | Neha Pillai         | SRN: R24EE001  | Sem: 6              | Grade: A                                           |
| [MECH]                 | Rohan Mehta         | SRN: R24ME001  | Sem: 7              | Grade: B                                           |
| [MECH]                 | Vivek Kumar         | SRN: R24ME003  | Sem: 5              | Grade: C                                           |
| [AHS ]                 | Shreya Joshi        | SRN: R24AH001  | Sem: 4              | Grade: A                                           |
| [AHS ]                 | Tanvi Deshmukh      | SRN: R24AH003  | Sem: 2              | Grade: B                                           |
| [ECE ]                 | Ananya Krishnan     | SRN: R24EC003  | Sem: 3              | Grade: O                                           |
| [ISE ]                 | Kavya Rao           | SRN: R24IS002  | Sem: 5              | Grade: A                                           |
| [CSE ]                 | Priya Sharma        | SRN: R24CS002  | Sem: 5              | Grade: O                                           |
| [ECE ]                 | Sneha Iyer          | SRN: R24EC001  | Sem: 5              | Grade: A                                           |
| FACULTY (22 Appointed) |                     |                |                     |                                                    |
| [CSE ]                 | Dr. Vikram Shah     | ID: FAC-CSE-01 | Professor & HOD     | Spec: Artificial Intelligence & Machine Learning   |
| [CSE ]                 | Prof. Ananya Rao    | ID: FAC-CSE-02 | Assistant Professor | Spec: Cloud Computing & Distributed Systems        |
| [ECE ]                 | Dr. Rajiv Nair      | ID: FAC-ECE-01 | Professor & Dean    | Spec: VLSI Design & Embedded Systems               |
| [ECE ]                 | Prof. Meera Iyer    | ID: FAC-ECE-02 | Associate Professor | Spec: Wireless Communication & Signal Processing   |
| [MECH]                 | Dr. Suresh Bhat     | ID: FAC-MEC-01 | Professor           | Spec: Thermal Engineering & Fluid Dynamics         |
| [MECH]                 | Prof. Kavya Menon   | ID: FAC-MEC-02 | Associate Professor | Spec: Robotics & Computer Integrated Manufacturing |
| [CIVIL]                | Dr. Arvind Kumar    | ID: FAC-CIV-01 | Professor & HOD     | Spec: Structural Engineering & Earthquake Dynamics |
| [CIVIL]                | Prof. Neha Sharma   | ID: FAC-CIV-02 | Assistant Professor | Spec: Environmental Geotechnics                    |
| [ISE ]                 | Dr. Harish Kulkarni | ID: FAC-ISE-01 | Professor           | Spec: Cybersecurity & Cryptography                 |

## Personnel Directory

|                                            |        |                              |                |                      |                                                     |
|--------------------------------------------|--------|------------------------------|----------------|----------------------|-----------------------------------------------------|
| Faculty<br>Members<br>by<br>Department     | [ISE ] | Dr. Harish Kulkarni          | ID: FAC-ISE-01 | Professor            | Spec: Cybersecurity & Cryptography                  |
|                                            | [ISE ] | Prof. Sunita Reddy           | ID: FAC-ISE-02 | Associate Professor  | Spec: Big Data Analytics & Data Science             |
|                                            | [EEE ] | Dr. Ramesh Murthy            | ID: FAC-EEE-01 | Professor & HOD      | Spec: Renewable Energy Systems & Smart Grids        |
|                                            | [EEE ] | Prof. Deepa Joseph           | ID: FAC-EEE-02 | Assistant Professor  | Spec: Power Electronics & Electric Drives           |
|                                            | [BT ]  | Dr. Sandeep Hegde            | ID: FAC-BIO-01 | Professor            | Spec: Genetic Engineering & Molecular Biology       |
|                                            | [BT ]  | Prof. Rashmi Prabhu          | ID: FAC-BIO-02 | Associate Professor  | Spec: Bioinformatics & Bioprocess Technology        |
|                                            | [AHS ] | Dr. Anand Swamy              | ID: FAC-AHS-01 | Professor & Dean     | Spec: Clinical Pathology & Diagnostic Medicine      |
|                                            | [AHS ] | Prof. Shilpa Shenoy          | ID: FAC-AHS-02 | Associate Professor  | Spec: Medical Imaging & Healthcare Informatics      |
|                                            | [COM ] | Dr. Bhaskar Shenoy           | ID: FAC-COM-01 | Professor & Director | Spec: Corporate Finance & Taxation                  |
|                                            | [COM ] | Prof. Lakshmi Narayanan      | ID: FAC-COM-02 | Associate Professor  | Spec: Financial Accounting & Auditing               |
| Faculty<br>Members<br>by<br>Specialization | [MGT ] | Dr. Keshava Prasad           | ID: FAC-MGT-01 | Professor & Dean     | Spec: Strategic Management & International Business |
|                                            | [MGT ] | Prof. Monica D'Souza         | ID: FAC-MGT-02 | Associate Professor  | Spec: Organizational Behavior & Human Resources     |
|                                            | [PA ]  | Dr. Vasundhara Das           | ID: FAC-ART-01 | Professor & HOD      | Spec: Classical Indian Music & Ethnomusicology      |
|                                            | [PA ]  | Prof. Shankar Mahadevan Nair | ID: FAC-ART-02 | Associate Professor  | Spec: Theatrical Arts & Choreography                |
| =====                                      |        |                              |                |                      |                                                     |

```
=====
Enter your choice: 10
```

```
=====
 OOP & JAVA EVALUATOR MODE
=====
```

1. Data Types, Variables & Scope
2. Operators & Operator Precedence
3. Type Conversion & Casting
4. Control Flow & Jump Statements
5. Arrays & String Methods
6. Constructors & Method Overloading
7. Static Members & Parameter Passing
8. Inheritance, super & final
9. Polymorphism & Dynamic Binding
10. Abstract Classes & Methods
11. Interfaces & Interface References
12. Object Class Methods
13. Packages & CLASSPATH
14. Run All Syllabus Demonstrations
0. Back to Main Menu

```
=====
Enter your choice: 1
```

```
=====
1. DATA TYPES, VARIABLES & SCOPE DEMONSTRATION
=====
```

Comprehensive coverage of Java 8 primitive data types:

1. byte (8-bit) : semester = 5
2. short (16-bit) : credits = 24
3. int (32-bit) : age = 20
4. long (64-bit) : admissionSequence = 2026000001
5. float (32-bit) : scholarship = 15.0%
6. double (64-bit) : fee = INR 105315.00
7. char (16-bit) : grade = 'S'
8. boolean (1-bit) : hosteller = true

4. long (64-bit) : admissionSequence = 2026000001
5. float (32-bit) : scholarship = 15.0%
6. double (64-bit) : fee = INR 105315.00
7. char (16-bit) : grade = 'S'
8. boolean (1-bit) : hosteller = true

Variable Scope Demonstration:

- Static Scope : Shared at class level (e.g., StudentService.MAX\_STUDENTS = 100)
- Instance Scope : Tied to object lifecycle (e.g., this.name, this.feeDue)
- Local Scope : Confined to enclosing method/block (e.g., local method variables)

=====

=====

OOP & JAVA EVALUATOR MODE

=====

1. Data Types, Variables & Scope
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9. Polymorphism & Dynamic Binding
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12. Object Class Methods
13. Packages & CLASSPATH
14. Run All Syllabus Demonstrations
0. Back to Main Menu

=====

Enter your choice: 5

=====

5. ARRAYS & STRING METHODS DEMONSTRATION

=====

```

=====
1. Arrays in SMS:
 - Primitive array : int[] marks = new int[5] (5 core academic subject marks)
 - Object array : Student[] students = new Student[100] (in-memory storage)
 - Bubble Sort : In-place lexicographical swapping on Student references
 - Array Shift : Left-shifting elements on deletion to maintain contiguity
 - Defensive Copy : Arrays.copyOf() ensuring encapsulation integrity

2. java.lang.String Class Exploration:

=====
DEMONSTRATION: Exploring java.lang.String Class Methods
=====
Original String: " REVA University Student Management System "
1. length() : 45
2. toUpperCase() : REVA UNIVERSITY STUDENT MANAGEMENT SYSTEM
3. toLowerCase() : reva university student management system
4. trim() : "REVA University Student Management System"
5. contains("REVA") : true
6. substring(0, 4) : RE
7. equalsIgnoreCase("reva"): false
=====

=====
OOP & JAVA EVALUATOR MODE
=====

1. Data Types, Variables & Scope
2. Operators & Operator Precedence
3. Type Conversion & Casting
4. Control Flow & Jump Statements
5. Arrays & String Methods
6. Constructors & Method Overloading
7. Static Members & Parameter Passing
8. Inheritance, super & final
9. Polymorphism & Dynamic Binding
10. Abstract Classes & Methods
11. Interfaces & Interface References

```

OOP Evaluator Demonstrations

## How to Run

```

run.bat
javac -d bin -sourcepath src src/com/reva/sms/model/*.java src/com/reva/sms/service/*.java src
java -cp bin com.reva.sms.main.Main
Get-Content test_input.txt | java -cp bin com.reva.sms.main.Main

```

# Project Structure

---

```
JAVA/src/com/reva/sms/
 main/ Main.java # Menu, input validation, demo dispatch
 model/ Department.java # Enum: 11 departments, credit rates
 Faculty.java # Person subclass
 Payable.java # Interface extending Reportable
 Person.java # Abstract base (id, name, age)
 Reportable.java # Interface, default getReportHeader()
 Student.java # Subclass implementing Payable
 service/ FeeCalculator.java # Fee computation, operator demo
 GradeCalculator.java # final utility class, grading
 StudentService.java # CRUD, sort, search, analytics
 bin/ docs/ run.bat test_input.txt sample_output.txt
```

Hierarchy: Reportable ← Payable ← Student ; Person ← Student / Faculty

## Syllabus Coverage — Unit I & II (21 Items)

---

| #  | Topic                                                | File(s)                                                                     | Evidence                                                                                                                                   |
|----|------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Encapsulation & Classes                              | see <code>src/</code>                                                       | Private fields, validated getters/setters                                                                                                  |
| 2  | Data Types, Scope & Constants                        | <code>Student</code> , <code>Person</code>                                  | All 8 primitives; static/instance/local                                                                                                    |
| 3  | Operators & Precedence                               | <code>FeeCalculator</code> ,<br><code>Student</code>                        | <code>demonstrateOperatorPrecedence()</code> ; <code>&lt;&lt;</code> <code>\ =</code> &                                                    |
| 4  | Type Conversion & Casting                            | <code>GradeCalculator</code> ,<br><code>Person</code>                       | <code>(double)total</code> ; <code>short→double</code> ; <code>(Person)obj</code>                                                          |
| 5  | Enumerated Types                                     | <code>Department</code>                                                     | 11-constant enum, <code>fromString()</code>                                                                                                |
| 6  | Control Flow & Jumps                                 | <code>Main</code> ,<br><code>GradeCalculator</code>                         | <code>if-else-if</code> , <code>switch</code> , <code>while</code> , <code>do-while</code> ,<br><code>break</code> / <code>continue</code> |
| 7  | Arrays                                               | <code>Student</code> ,<br><code>StudentService</code>                       | <code>int[]marks</code> ; bubble sort; left-shift delete                                                                                   |
| 8  | Console I/O & Formatting                             | <code>Main</code> , <code>StudentService</code>                             | <code>Scanner</code> ; <code>printf</code> tabular reports                                                                                 |
| 9  | Constructors & <code>this</code>                     | <code>Person</code> , <code>Student</code>                                  | Overloaded constructors; <code>this(...)</code> chaining                                                                                   |
| 10 | Method Overloading                                   | <code>GradeCalculator</code> ,<br><code>FeeCalculator</code>                | <code>calculateGrade</code> ×3; <code>calculateSemesterFee</code> ×3;<br><code>searchStudents</code> ×2                                    |
| 11 | Parameter Passing                                    | <code>StudentService</code>                                                 | <code>demonstrateParameterPassing()</code> : primitive vs reference                                                                        |
| 12 | Static Members                                       | <code>Person</code> , <code>Student</code> ,<br><code>StudentService</code> | <code>generateNextIdSafe()</code> ; counters; constants                                                                                    |
| 13 | Access Modifiers                                     | all files                                                                   | <code>public</code> , <code>protected</code> ( <code>name</code> , <code>age</code> ), <code>private</code>                                |
| 14 | <code>String</code> API                              | <code>StudentService</code> ,<br><code>Department</code>                    | <code>trim()</code> , <code>contains()</code> , <code>equalsIgnoreCase()</code>                                                            |
| 15 | Inheritance, <code>super</code> & <code>final</code> | <code>Person</code> ,<br><code>GradeCalculator</code>                       | <code>super(...)</code> ; <code>final class</code> ; <code>final getId()</code>                                                            |
| 16 | <code>Object</code> Class Overrides                  | <code>Person</code> , <code>Student</code>                                  | <code>toString()</code> , SRN-keyed <code>equals()</code> ,<br><code>hashCode()</code>                                                     |
| 17 | Dynamic Binding                                      | <code>StudentService</code>                                                 | <code>demonstratePolymorphicDispatch()</code> ;<br><code>instanceof</code> pattern                                                         |
| 18 | Abstract Classes                                     | <code>Person</code>                                                         | <code>abstract class Person</code> ; <code>abstract displayDetails()</code>                                                                |

| #  | Topic                        | File(s)                                        | Evidence                                                     |
|----|------------------------------|------------------------------------------------|--------------------------------------------------------------|
| 19 | Interfaces & Default Methods | <code>Reportable</code> , <code>Student</code> | <code>default getReportHeader() ; Payable p = student</code> |
| 20 | Interface Inheritance        | <code>Payable</code>                           | <code>interface Payable extends Reportable</code>            |
| 21 | Packages & CLASSPATH         | <code>com.reva.sms.*</code>                    | <code>main / model / service ; -d bin / -cp bin</code>       |

## CLASSPATH

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`javac -d bin` writes `.class` files preserving package dirs; `java -cp bin` resolves `com.reva.sms.main.Main` from that root.